



Recitation 7

Yuchao Chen, Mingtao Huang

{cyc20, huangmt19}@mails.tsinghua.edu.cn

Tsinghua University

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3. An image is partitioned into two regions, one white and the other black. A reading taken from a randomly chosen point in the white section will be normally distributed with $\mu = 4$ and $\sigma^2 = 4$, whereas one taken from a randomly chosen point in the black region will have a normally distributed reading with parameters $(6, 9)$. A point is randomly chosen on the image and has a reading of 5. If the fraction of the image that is black is α , for what value of α would the probability of making an error be the same, regardless of whether one concluded that the point was in the black region or in the white region?

3. An image is partitioned into two regions, one white and the other black. A reading taken from a randomly chosen point in the white section will be normally distributed with $\mu = 4$ and $\sigma^2 = 4$, whereas one taken from a randomly chosen point in the black region will have a normally distributed reading with parameters $(6, 9)$. A point is randomly chosen on the image and has a reading of 5. If the fraction of the image that is black is α , for what value of α would the probability of making an error be the same, regardless of whether one concluded that the point was in the black region or in the white region?

We want to find α such that we cannot estimate the color of the chose point in a “more correct way”. That is to say, given the reading is 5, the probability of the point being in the black region is the same as being in the white region.

$$\mathbf{P}(\text{black} \mid \text{reading is } 5) = \mathbf{P}(\text{white} \mid \text{reading is } 5) = 0.5$$

Applying Bayes' rule,

$$\begin{aligned}\mathbf{P}(\text{black} \mid 5) &= \frac{\mathbf{P}(\text{black and } 5)}{\mathbf{P}(5)} \\&= \frac{\mathbf{P}(5 \mid \text{black})\mathbf{P}(\text{black})}{\mathbf{P}(5 \mid \text{black})\mathbf{P}(\text{black}) + \mathbf{P}(5 \mid \text{white})\mathbf{P}(\text{white})} \\&= \frac{\frac{1}{3\sqrt{2\pi}}e^{-(5-6)^2/(2 \cdot 9)} \cdot \alpha}{\frac{1}{3\sqrt{2\pi}}e^{-(5-6)^2/(2 \cdot 9)} \cdot \alpha + \frac{1}{2\sqrt{2\pi}}e^{-(5-4)^2/(2 \cdot 4)} \cdot (1 - \alpha)} \\&= \frac{\frac{\alpha}{3}e^{-1/18}}{\frac{\alpha}{3}e^{-1/18} + \frac{1 - \alpha}{2}e^{-1/8}} = \frac{1}{2} \\&\quad \alpha = \frac{3e^{-1/8}}{3e^{-1/8} + 2e^{-1/18}} \approx 0.583\end{aligned}$$

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- Joint PDF:

$$\mathbf{P}((X, Y) \in S) = \iint_S f_{X,Y}(x, y) dx dy$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

- Marginal PDF:

$$\mathbf{P}(X \in A) = \mathbf{P}(X \in A, Y \in (-\infty, \infty)) = \int_A \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

- Joint CDF:

$$F_{X,Y}(x, y) = \mathbf{P}(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) ds dt$$

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x, y)$$

- Conditioning

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \quad \text{if } f_Y(y) > 0$$

- The continuous Bayes' rule

$$f_{X|Y}(x | y) = \frac{f_X(x)f_{Y|X}(y | x)}{f_Y(y)} = \frac{f_X(x)f_{Y|X}(y | x)}{\int_{-\infty}^{\infty} f_X(t)f_{Y|X}(y | t) dt}$$

- Expectation:

$$\mathbf{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy$$

If g is linear, of the form $aX + bY + c$,

$$\mathbf{E}[aX + bY + c] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c$$

- **Total expectation theorem:**

$$\mathbf{E}[X] = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{E}[X | A_i]$$

$$\mathbf{E}[X] = \mathbf{E}_Y [\mathbf{E}_{X|Y} [X | Y]] = \int_{-\infty}^{\infty} \mathbf{E}[X | Y = y] f_Y(y) dy$$

- Independence:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \quad \text{for all } x, y$$

$$f_{X|Y}(x | y) = f_X(x), \quad \text{for all } y \text{ with } f_Y(y) > 0 \text{ and all } x$$

- If X and Y are independent, then

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)] \quad \forall g, h$$

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

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- A light bulb is known to have an exponentially distributed lifetime Y . The parameter λ of the pdf is uniformly distributed in the interval $[1, 3/2]$. We test a light bulb and record its lifetime. What can we say about the underlying λ ?



Solution:

- We model λ in terms of a uniform RV Λ with PDF

$$f_{\Lambda}(\lambda) = 2, \quad \text{for } 1 \leq \lambda \leq \frac{3}{2}.$$

$$f_{Y|\Lambda}(y | \lambda) = \lambda e^{-\lambda y}$$

- The available information about Λ is captured by the conditional pdf $f_{\Lambda|Y}(\lambda | y)$

$$\begin{aligned} f_{\Lambda|Y}(\lambda | y) &= \frac{f_{\Lambda}(\lambda) f_{Y|\Lambda}(y | \lambda)}{\int_{-\infty}^{\infty} f_{\Lambda}(t) f_{Y|\Lambda}(y | t) dt} \\ &= \frac{2\lambda e^{-\lambda y}}{\int_1^{3/2} 2t e^{-ty} dt}, \quad \text{for } 1 \leq \lambda \leq \frac{3}{2}. \end{aligned}$$

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1. Let X be an exponential random variable with parameter $\lambda > 0$. Calculate the probability that X belongs to one of the intervals $[n, n + 1]$ with n odd.

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Let A denotes the event that X belongs to one of the intervals $[n, n + 1]$ with n odd

$$\begin{aligned}\mathbf{P}(n \leq X \leq n + 1) &= F_X(n + 1) - F_X(n) \\ &= e^{-\lambda n}(1 - e^{-\lambda})\end{aligned}$$

Since the intervals are **disjoint**, the probability of the event A is

$$\begin{aligned}\mathbf{P}(A) &= \sum_{k=0}^{\infty} e^{-\lambda(2k+1)}(1 - e^{-\lambda}) \\ &= \frac{e^{-\lambda}}{1 + e^{-\lambda}}.\end{aligned}$$

2. Exponential Random Variable is Memoryless. The time T until a new light bulb burns out is an exponential random variable with parameter λ . Ariadne turns the light on, leaves the room, and when she returns, t time units later, finds that the bulb is still on, which corresponds to the event $A = \{T > t\}$. Let X be the additional time until the bulb burns out. What is the conditional CDF of X , given the event A ?

2. Exponential Random Variable is Memoryless. The time T until a new light bulb burns out is an exponential random variable with parameter λ . Ariadne turns the light on, leaves the room, and when she returns, t time units later, finds that the bulb is still on, which corresponds to the event $A = \{T > t\}$. Let X be the additional time until the bulb burns out. What is the conditional CDF of X , given the event A ?

$$\begin{aligned}\mathbf{P}(X > x|A) &= \mathbf{P}(T > t + x|T > t) \\ &= \frac{\mathbf{P}(T > t + x)}{\mathbf{P}(T > t)} \\ &= e^{-\lambda x}.\end{aligned}$$

3. Let the random variables X and Y have a joint pdf which is uniform over the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.
- (a) Find the joint pdf of X and Y .
 - (b) Find the marginal pdf of Y .
 - (c) Find the conditional pdf of X given Y .
 - (d) Find $\mathbf{E}[X|Y = y]$, and use the total expectation theorem to find $\mathbf{E}[X]$ in terms of $\mathbf{E}[Y]$.
 - (e) Use the symmetry of the problem to find the value of $\mathbf{E}[X]$.

(a)

$$f_{X,Y}(x,y) = \begin{cases} 2 & \text{if } y \leq 1-x, 0 \leq y \leq 1, 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(b)

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx = \int_0^{1-y} 2 dx = 2(1-y), \quad 0 \leq y \leq 1.$$

(c)

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{1}{1-y}, \quad 0 \leq x \leq 1-y.$$

(d) For $0 \leq y \leq 1$

$$\mathbf{E}[X|Y = y] = \int_0^{1-y} x \frac{1}{1-y} dx = \frac{1-y}{2}.$$

The total expectation theorem yields

$$\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X|Y]] = \int_0^1 \mathbf{E}[X|Y = y] f_Y(y) dy = \frac{1 - \mathbf{E}[Y]}{2}.$$

(e) Because of the symmetry, we must have $\mathbf{E}[X] = \mathbf{E}[Y]$. Therefore, $\mathbf{E}[X] = 1/3$.

4. We have a stick of unit length, and we break it into three pieces. We choose randomly and independently two points on the stick using a uniform pdf, and we break the stick at these points. What is the probability that the three pieces we are left with can form a triangle?



4. We have a stick of unit length, and we break it into three pieces. We choose randomly and independently two points on the stick using a uniform pdf, and we break the stick at these points. What is the probability that the three pieces we are left with can form a triangle?

Denote the position of the first break by X and the position of the second break by Y . We assume $X < Y$, then the three pieces have length X , $Y - X$, and $1 - Y$. They can form a triangle if

$$X < Y - X + 1 - Y, \quad 1 - Y < X + Y - X, \quad Y - X < X + 1 - Y$$

These conditions simplify to

$$X < 0.5, \quad Y > 0.5, \quad X > Y - 0.5$$

Thus the probability of the event that the three pieces form a triangle and $X < Y$ is

$$\begin{aligned}\mathbf{P} &= \int_{1/2}^1 \int_{y-1/2}^{1/2} f_{X,Y}(x,y) dx dy \\ &= \int_{1/2}^1 \int_{y-1/2}^{1/2} 1 dx dy \\ &= \frac{1}{8}\end{aligned}$$

By symmetry, when $X > Y$, $\mathbf{P} = \frac{1}{8}$. Therefore, the probability of the three pieces form a triangle is $1/4$.